

Editorial

Cellular and Molecular Mechanisms of Disease: A 5-Year Cycle in the Long Life of *The American Journal of Pathology*

This issue of *The American Journal of Pathology* marks the completion of 5 years since the American Society for Investigative Pathology (ASIP), the publisher of the *Journal*, assumed full responsibility for its management. The shift to "self" publication brought with it many changes and uncertainties. However, the main goal of the ASIP and the new editorial team at the *Journal* was clear: to make it the premier journal dedicated to cellular and molecular mechanisms of disease. At the completion of this 5-year cycle it is thus appropriate to ask whether this goal has been achieved and to look at some important issues and trends related to research in disease pathogenesis.

At a time in which new journals in biology and medicine are launched almost every month, it is a matter of great pride for us that the *Journal* has been in existence under its present name for 72 years, and that its origins under other names can be traced back to the turn of the century. But this century of history has meant much more than a succession of different publication formats finally leading to the present journal. It actually reflects the profound changes in the scope and role of pathology in biomedical research and in the practice of medicine during this period. One hundred years ago, pathology was *the* scientific discipline of medicine. Relying on the autopsy and experiments in animals and microorganisms, pathologists discovered, described, and attempted to explain the pathogenesis of the most important human diseases. However, by the middle of the century many pathology departments in academic institutions were becoming specialty departments focused on morphological diagnosis from autopsy and surgical specimens. The separation between basic research and surgical pathology diagnosis became accentuated as surgical pathology grew as a subspecialty, and the biological basis

of medical research began an era of remarkable expansion. The origins and evolution of this dichotomy are reviewed in an excellent article by Juan Rosai published in this issue of the *Journal*.

Despite these historical trends, academic pathologists today are once again in a position of being the physician-scientists of medicine. In the larger arena of biomedical research, the expertise of a well trained research pathologist has become invaluable for studies of disease pathogenesis that rely on transgenic and knockout animal models. Right now and even more so in the future, individuals trained both in morphological pathology and in modern basic science approaches to disease pathogenesis (including, among other fields, molecular biology, genetics, and immunology) function not only as the "interpreters" of disease for the scientific community but also as creative scientists who develop and apply new methods of disease diagnosis. A good example of what is ahead is the use of DNA microarrays ("chips") for genetic diagnosis. While the technical aspect of this methodology is being developed by biotechnology experts, the actual application of the technique and its interpretation as a diagnostic tool will most certainly become the responsibility of pathologists trained in both diagnosis and research. Similarly, the sequencing of the human genome now demands the determination of the pathological consequences, if any, of the multitude of mutations and genetic polymorphisms that are being detected in ever-increasing numbers. Again, physicians and scientists trained in diagnostic and investigative pathology are the ones best qualified to undertake this task.

It is to the community of scientists interested in disease pathogenesis, located in pathology departments as well as in other basic science and clinical units, that the *Journal* is directed. What are the sta-

tistical highlights for the 1992–1997 period? Judging from the number of submissions, the quality of the material published, and the citation index ranking (*The American Journal of Pathology* is the number 1 peer-reviewed journal in the field of pathology), the last 5 years have been a great success. Submissions have increased by 85% since 1992. During the last 3 years, submissions increased by 11% annually. The *Journal* continues to be highly selective, accepting 20 to 25% of newly submitted manuscripts. Final decisions to accept or reject manuscripts are made by the Editor-in-Chief or the Senior Editor, based on the recommendations of two external reviewers and an Associate Editor. About 2% of manuscripts are accepted without requiring further change, but the vast majority of acceptable manuscripts are conditionally accepted, pending revisions. Meritorious manuscripts that may require substantial revisions are not considered as acceptable. Despite the increased load of manuscripts, the average time for review of regular manuscripts in 1996 was 48 days. In May 1996 we established procedures to expedite the review of Short Communications. Since that date, the review of Short Communications has taken an average of 23 days. The time between final acceptance of a manuscript and its publication has also been shortened. We are approaching our objective of publishing manuscripts 8 weeks after their final acceptance, a goal that will be reached on or before January 1998. The changes in the efficiency of the review and publication processes have led to a significant decrease in the time from first receipt of a manuscript to its publication (this includes the initial review and letter of disposition, submission of a revised manuscript and final letter of disposition, and the time interval to publication). In 1996 it took an average of 190 days from the date of receipt of a manuscript to its printing, as compared to 293 days in 1993. Another interesting and pleasing aspect of the *Journal* is that although it is called "American," it has become an international journal. During the past year, 47% of published manuscripts originated in foreign countries. The countries most often represented were Japan, the United Kingdom, Italy, The Netherlands, and Canada.

The numbers presented above speak to the efficiency of the process. However, they do not address the important issues of quality and fairness. Our

philosophy is that quality and fairness are achieved only when a large and diverse group of reviewers participate in the process. We also insist that the reviewers' comments be "reviewed" by Associate Editors before the Editor-in-Chief issues a letter of disposition. We have appointed Associate Editors who are experts in their field and who are geographically dispersed so as to avoid institutional bias. In 1997, the number of Associate Editors was increased to accommodate the larger number of manuscripts received by the *Journal*. We have had a constant input of new ideas and approaches by instituting rotating memberships for Associate Editors and members of the Editorial Board. We sincerely thank retiring Associate Editors for their outstanding performance and dedication, and we salute the new group. Likewise, we thank the members of the Editorial Board and the more than 1,000 reviewers who have helped the *Journal* achieve its present level of quality.

I now return to the question of whether the *Journal* has reached the goal of becoming the premier journal dedicated to cellular and molecular mechanisms of disease. As much as we scientists like to question our own views, I would like to say that my answer to the question is a confident "yes." However, I couple the "yes" with a "but" to indicate that quality, efficiency, and fairness are moving targets that become even more difficult to reach when one is approaching them. Of course not everything is rosy. We, as others, make mistakes, although we work hard to avoid them. From an Editor's perspective, I am disappointed by the rising number of overlapping manuscripts and, on a few occasions, outright duplications. Submission of Minimum Publishable Units as well as problems of conflict of interest continue to detract from the quality and free dissemination of research. Nevertheless, our goal for the *Journal* is being realized, giving us great confidence in its future. Research in disease pathogenesis is in a period of enormous excitement, creativity, and expansion. With the help of scientists and physicians as authors and reviewers, the *Journal* will continue to play a major role in encouraging and publishing high quality work aimed at understanding cellular and molecular mechanisms of disease.

Nelson Fausto
Editor-in-Chief